Math Preliminaries

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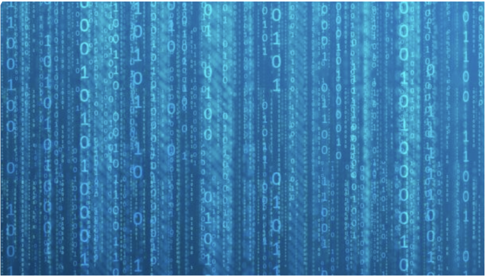

Linear Algebra

Linear Algebra

- In Data Mining, we will represent data points using vectors or ordered sets of coordinates (corresponding to various attributes/features).
- The branch of mathematics that concerns with such coordinated representations is called **linear algebra**.
- Reference: Chapter 02 of the MML book [https://mml-book.github.io/book/mml-book.pdf]

Vectors

- We will denote a vectors by $\overline{X}, \overline{Y} \ldots$(uppercase letters with a bar).
- We will use column vectors throughout this module (transposed by T when written as row vectors) e.g.
  $\overline{X} = (3, 2, -1.5)^T$
- $\overline{X} \in R^d$means that is $\overline{X}$ is a d dimensional vector with real coordinates

Matrices

- We obtain matrices by arranging a collection of vectors by columns or rows.
- Similarly to the vectors, we use uppercase letters with a bar to denote matrices such as $\overline{M}$
- $\overline{M} \in \mathbb{R}^{n \times m}$ means that $\overline{M}$ is a matrix with n rows and m columns
- When $n = m$ we say M is **square**
- We denote the (i, j) element of $\overline{M}$ by $\overline{M}_{i,j}$
- If $\overline{M}_{i,j} = \overline{M}_{j,i}$ for all i and j, we say $\overline{M}$ is **symmetric**. Otherwise, $\overline{M}$ is asymmetric.

Vector Arithmetic

- Given two vectors $\overline{X}, \overline{Y} \in \mathbb{R}^d$, where
$$\overline{X} = (x_1, \ldots, x_d)^T \quad \text{and} \quad \overline{Y} = (y_1, \ldots, y_d)^T$$
- Their **addition** is given by the vector
$$\overline{Z} = (z_1, \ldots, z_d)^T$$
where the i-th element z_i is given by
$$z_i = x_i + y_i$$
- Their **inner-product** (also known as **dot product**) is defined as
$$\overline{X}^T \overline{Y} = \sum_{i=1}^{d} x_i y_i$$
- Their **outer-product** $\overline{X}\,\overline{Y}^T$ is defined as the matrix $\overline{M} \in \mathbb{R}^{d \times d}$, where

Matrix Arithmetic

- Matrices of the same shape (number of rows and columns) can be **added** element-wise
$$\overline{A} + \overline{B} = \overline{C}, \quad \text{where } \overline{C}_{i,j} = \overline{A}_{i,j} + \overline{B}_{i,j}$$
- Matrices can be **multiplied** if the number of columns of the first matrix is equal to the number of rows of the second matrix. Let $\overline{A} \in \mathbb{R}^{n\times m}$ and $\overline{B} \in \mathbb{R}^{m\times d}$, then the matrix $\overline{C} = \overline{A}\,\overline{B}$ has n rows and d columns,

$$\overline{C}_{i,j} = \sum_{k=1}^{m} \overline{A}_{i,k}\,\overline{B}_{k,j}$$

Transpose and Inverse

- The **transpose** of a matrix $\overline{A} \in \mathbb{R}^{n \times d}$ is denoted by $\overline{A}^T$ and is a matrix from $\mathbb{R}^{d \times n}$, where
$$\overline{A}^T_{i,k} = \overline{A}_{k,i}$$
- $$(\overline{AB})^T = \overline{B}^T \overline{A}^T$$
- The **inverse** of a square matrix $\overline{A} \in \mathbb{R}^{n \times n}$ is denoted by $\overline{A}^{-1}$ and satisfies
$$\overline{A}\,\overline{A}^{-1} = \overline{A}^{-1}\overline{A} = I,$$
where $I \in \mathbb{R}^{n \times n}$ is the **unit matrix** (all diagonal elements are set to 1 and non-diagonal elements are set to 0)

Linear Independence

- Let us consider a vector $\overline{V}$ formed as the linearly-weighted sum of a set of vectors $\{\overline{X}_1, \ldots, \overline{X}_k\}$ with respective coefficients $\lambda_1, \ldots, \lambda_k$ as follows:

$$\overline{V} = \lambda_1 \overline{X}_1 + \cdots + \lambda_k \overline{X}_k = \sum_{i=1}^{k} \lambda_i \overline{X}_i$$

- $\overline{V}$ is called a **linear combination** of $\overline{X}_1, \ldots, \overline{X}_k$
- Vectors $\overline{X}_1, \ldots, \overline{X}_k$ are called **linearly dependent** if there exist $\lambda_1, \ldots, \lambda_k$, not all zero, such that

$$\overline{0} = \lambda_1 \overline{X}_1 + \cdots + \lambda_k \overline{X}_k$$

- Otherwise $\overline{X}_1, \ldots, \overline{X}_k$ are called **linearly independent**

Rank

- The number of linear independent columns of a matrix $\overline{A} \in \mathbb{R}^{m \times n}$ $(m \leq n)$ equals the number of linearly independent rows and is called the **rank** of $\overline{A}$, denoted by $\mathrm{rank}(\overline{A})$.
- $\mathrm{rank}(\overline{A}) \leq \min\{m, n\} = m$
- If $\mathrm{rank}(\overline{A}) = m$, then $\overline{A}$ is said to be **full-rank**, otherwise **rank-deficient**.
- Only full-rank square matrices are invertible.

Matrix Trace

The sum of diagonal elements is called the trace of the matrix. Specifically,

$$\mathrm{tr}(\overline{A}) = \sum_i \overline{A}_{i,i}$$

Example

$$\overline{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \mathrm{tr}(\overline{A}) = 2$$

Eigenvalues and Eigenvectors

Let $\overline{A} \in \mathbb{R}^{n \times n}$ be a square matrix.

A non-zero vector $\overline{X} \in \mathbb{R}^n$ is an **eigenvector** of $\overline{A}$ if

$$\overline{A}\,\overline{X} = \lambda \overline{X}$$

for some $\lambda \in \mathbb{R}$, which is called the **eigenvalue** of $\overline{A}$ corresponding to $\overline{X}$.

Differential Calculas

Derivatives of basic functions

$$\frac{d}{dx}a = 0, \quad \text{where } a \text{ is a constant (i.e., does not depend on } x\text{)}$$

$$\frac{d}{dx}x^a = a \cdot x^{a-1}$$

$$\frac{d}{dx}e^x = e^x, \quad \text{where } e \approx 2.71 \text{ is Euler's number}$$

$$\frac{d}{dx}\log(x) = \frac{1}{x}, \quad \text{where } x > 0$$

$$\frac{d}{dx}\sin(x) = \cos(x)$$

$$\frac{d}{dx}\cos(x) = -\sin(x)$$

Note: $\frac{d}{dx}f(x)$ is the same as $f'(x)$

Differentiation rules

Sum rule:

$$(\alpha f + \beta g)' = \alpha f' + \beta g'$$

Product rule:

$$(fg)' = f'g + fg'$$

Quotient rule:

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

Chain rule: If $f(x) = h(g(x))$, then

$$f'(x) = h'(g(x)) \cdot g'(x) = \frac{d}{dg(x)}h \cdot \frac{d}{dx}g$$

Partial derivative

A **partial derivative** of a function of several variables is its derivative with respect to one of those variables, with the others held constant.

Example

$$f(x, y) = 5x + y^2$$

$$\frac{\partial f}{\partial x} = 5 \qquad \frac{\partial f}{\partial y} = 2y \qquad \nabla_{(x,y)} f = (5,\ 2y)^T$$

Reference

Chapter 6 of the MML book

https://mml-book.github.io/book/mml-book.pdf

Optimisation

Continuous optimisation

- Unconstrained optimisation
- Constrained optimisation

Reference: Chapter 7 of the MML book
https://mml-book.github.io/book/mml-book.pdf

Unconstrained optimisation: Gradient Descent method

Problem formulation

$$\text{find } \min_{\overline{X}} f(\overline{X})$$

where

1. $\overline{X} = (x_1, x_2, \ldots, x_d)$ and
2. $f : \mathbb{R}^d \to \mathbb{R}$
3. f is differentiable and we are unable to analytically find a solution in closed form.

Unconstrained optimisation: Gradient Descent method

Let $f(\overline{X}) = f(x_1, \ldots, x_d)$ be a function depending on d variables.

The **gradient** of $f(\overline{X})$ is the vector consisting of the partial derivatives of f:

$$\nabla_{\overline{X}} f = \frac{\partial f}{\partial \overline{X}} = \left(\frac{\partial f(\overline{X})}{\partial x_1}, \ \frac{\partial f(\overline{X})}{\partial x_2}, \ \ldots, \ \frac{\partial f(\overline{X})}{\partial x_d} \right)^T$$

Unconstrained optimisation: Gradient Descent method

The gradient $\nabla_{\overline{X}} f$ evaluated in a point $\overline{X}_0$ gives a vector that points in the direction of the steepest ascent.
Starting from point $\overline{X}_0$ the function f **decreases faster** if one moves from $\overline{X}_0$ in the direction of the **negative gradient** of f at $\overline{X}_0$, i.e., in the direction of the vector

$$-(\nabla_{\overline{X}} f)(\overline{X}_0)$$

This means that for a small step-size $\gamma \geq 0$ the value of the function in point

$$\overline{X}_1 = \overline{X}_0 - \gamma \cdot (\nabla_{\overline{X}} f)(\overline{X}_0)$$

is smaller than in the initial point $\overline{X}_0$, i.e.,

$$f(\overline{X}_1) \leq f(\overline{X}_0)$$

Unconstrained optimisation: Gradient Descent method

Algorithm for finding local minimum of $f(\overline{X}) = f(x_1, \ldots, x_d)$

1. Pick an initial point $\overline{X}_0$
2. Iterate according to

$$\overline{X}_{i+1} = \overline{X}_i - \gamma_i \cdot \left((\nabla_{\overline{X}} f)(\overline{X}_i)\right)$$

For suitable step-sizes $\gamma_1, \gamma_2, \ldots$, the sequence

$$f(\overline{X}_0) \geq f(\overline{X}_1) \geq \cdots$$

converges to a **local minimum**.
Moral: gradient of a function is a useful tool for finding local optimal points of a function and is widely used in data mining and machine learning.

Constrained optimisation: method of Lagrange multipliers

Problem formulation

$$\text{find } \min_{\overline{X}} f(\overline{X})$$

$$\text{Subject to} \quad g(\overline{X}) = 0$$

where

1. $\overline{X} = (x_1, x_2, \ldots, x_d)$ and
2. $f : \mathbb{R}^d \to \mathbb{R}$
3. $g : \mathbb{R}^d \to \mathbb{R}$
4. f is differentiable and we are unable to analytically find a solution in closed form

Constrained optimisation: method of Lagrange multipliers

Problem formulation

$$\text{find } \min_{\overline{X}} f(\overline{X})$$

$$\text{Subject to} \quad g(\overline{X}) = 0$$

In order to solve this problem:

1. Form the Lagrangian function

$$\mathcal{L}(\overline{X}, \lambda) = f(\overline{X}) - \lambda \cdot g(\overline{X})$$

2. Find all stationary points $(\overline{X}_0, \lambda_0)$ of $\mathcal{L}(\overline{X}, \lambda)$, i.e. those points for which all partial derivatives of $\mathcal{L}(\overline{X}, \lambda)$ are equal to 0, or equivalently

$$\nabla_{(\overline{X}, \lambda)} \mathcal{L} = \overline{0}$$

3. Examine stationary points to find among them a solution to the problem

Probability

Common discrete probability distributions

- **Bernoulli distribution**: models binary outcomes (coin flip).

$$P(X = \text{head}) = p \quad \text{and} \quad P(X = \text{tail}) = 1 - p$$

- **Generalised Bernoulli distribution**: models $k > 2$ outcomes (rolls of a k-sided die)

$$P(X = 1) = p_1,\ P(X = 2) = p_2,\ \ldots,\ P(X = k) = p_k \quad \text{such that} \quad \sum_{i=1}^{k} p_i = 1$$

Common discrete probability distributions

- **Binomial distribution**: models a sequence of multiple flips of a coin

$$P(\text{in } n \text{ flips there are exactly } k \text{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}$$

- **Multinomial distribution**: models a sequence of multiple rolls of a k-sided die for $k > 2$
  If there are n rolls and n_i is the number of times the die came up on side i, then the probability of this event is

$$\frac{n!}{\prod_{i=1}^{k} n_i!} \cdot \prod_{i=1}^{k} p_i^{n_i}$$